

Algorithm no-

Algorithm Name: **Linear Search**

Code:

```
#include<bits/stdc++.h>45
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    vector<int>A={6,8,6,9,10,1,5,7,12,15,100,52,40,47,50,55};
    int n=A.size();
    bool flag=true;
    for(int i:A)
    {
        cout<<i<<" ";
    }
    newl;
    int a;
    cin>>a;
    for(int i=0;i<n;i++)
    {
        if(A[i]==a)
        {
            cout<<i<<" ";newl;
            flag=false;
        }
    }
    if(flag)
    {
        cout<<"Not found\n";
    }
}
```

Output:

```
6 8 6 9 10 1 5 7 12 15 100 52 40 47 50 55
10
5
Press any key to continue . . . |
```

```
6 8 6 9 10 1 5 7 12 15 100 52 40 47 50 55
20
Not found
Press any key to continue . . .
```

Algorithm no-

Algorithm Name: **Binary Search**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    vector<int>A={6,8,6,9,10,1,5,7,12,15,100,52,40,47,50,55};
    int n=A.size();
    vector<pair<int,int>>ab(n);
    for(int i=0;i<n;i++)
    {
        ab[i].first=A[i];ab[i].second=i+1;
        cout<<A[i]<<" ";
    }
    newl;
    while(1)
    {
        int b=0;
        for(int i=0;i<n-1;i++)
        {
            if(ab[i].first>ab[i+1].first)
            {
                swap(ab[i],ab[i+1]);
            }
            else
            {
                b++;
            }
        }
        if(b==n-1)
        {
            break;
        }
    }
    int a;
    cin>>a;
    int lb=0,up=n;
    int mid=(lb+up)/2;
    while(1)
    {
        if(ab[mid].first<a)
        {
            lb=mid;
            mid=(lb+up)/2;
        }
        else if(ab[mid].first==a)
        {
            cout<<ab[mid].second;
            break;
        }
    }
}
```

```

    }
    else if(ab[mid].first>a)
    {
        up=mid;
        mid=(lb+up)/2;
    }
    else
    {
        cout<<"Not found\n";
        break;
    }
}
}
}

```

Output:

```

6 8 6 9 10 1 5 7 12 15 100 52 40 47 50 55
10
5
Press any key to continue . . .

```

Algorithm no-

Algorithm Name: **2D Array Summation-Matrix Summation**

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    int r,c;
    cin>>r>>c;
    int R[r+1][c+1];
    for(int i=1;i<=r;i++)
    {
        for(int j=1;j<=c;j++)
        {
            cin>>R[i][j];
        }
    }
    newl;
    int a,b;
    cin>>a>>b;
    if(a==r&&b==c)

```

```

{
    int B[a+1][b+1];
    for(int i=1;i<=r;i++)
    {
        for(int j=1;j<=c;j++)
        {
            cin>>B[i][j];
        }
    }
    int C[r+1][c+1];
    for(int i=1;i<=r;i++)
    {
        for(int j=1;j<=c;j++)
        {
            C[i][j]=R[i][j]+B[i][j];
        }
    }
    newl;
    for(int i=1;i<=r;i++)
    {
        for(int j=1;j<=c;j++)
        {
            cout<<C[i][j]<<" ";
        }
        newl;
    }
}
else
{
    cout<<"Matrix Summation Error";
}
}

```

Output:

```

2 2
1 2
3 4

2 2
5 6
7 8

6 8
10 12

Press any key to continue . . . |

```

Algorithm no-

Algorithm Name: **2D Array Multiplication-Matrix Multiplication**

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    int a,b;
    cin>>a>>b;
    int A[a+1][b+1];
    for(int i=1;i<=a;i++)
    {
        for(int j=1;j<=b;j++)
        {
            cin>>A[i][j];
        }
    }
    newl;newl;
    int c,d;
    cin>>c>>d;
    int B[c+1][d+1],R[a+1][d+1];
    if(b==c)
    {
        for(int i=1;i<=c;i++)
        {
            for(int j=1;j<=d;j++)
            {
                cin>>B[i][j];
                R[i][j]=0;
            }
        }
        for(int i=1;i<=a;i++)
        {
            for(int j=1;j<=d;j++)
            {
                for(int m=1;m<=b;m++)
                {
                    R[i][j]=R[i][j]+A[i][m]*B[m][j];
                }
            }
        }
        newl;
        for(int i=1;i<=a;i++)
        {
            for(int j=1;j<=d;j++)
            {
                cout<<R[i][j]<<" ";
            }
            newl;
        }
    }
```

```

    }
    else
    {
        printf("Matrix Multiplication error");
    }
}

```

Output:

```

2 2
1 2
3 4

2 1
5
6

17
39

Press any key to continue . . .

```

Problem No- Round 892 (DIV 2.A)

Problem Name: **United We Stand**

Problem Link: <https://codeforces.com/contest/1859/problem/A>

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define newl cout<<endl
int main()
{
    _easy_
    int t;
    cin>>t;
    while(t--)
    {
        int n;
        cin>>n;
        vector<int>A(n);
        map<int,int>mp;
        int mx=-100;
        for(int i=0;i<n;i++)
        {

```

```

        cin>>A[i];
        mx=max(mx,A[i]);
        mp[A[i]]++;
    }
    if(mp[mx]==n)
    {
        cout<<-1;
        newl;
        continue;
    }
    cout<<n-mp[mx]<<" "<<mp[mx]<<endl;
    for(int i=0;i<n;i++)
    {
        if(A[i]!=mx)
        {
            cout<<A[i]<<" ";
        }
    }
    newl;
    for(int i=1;i<=mp[mx];i++)
    {
        cout<<mx<<" ";
    }
    newl;
}
}

```

Output:

218520036	Aug/12/2023 20:56 ^{UTC+6}	Arsenic33	A - United We Stand	GNU C++20 (64)	Accepted	30 ms	0 KB
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Problem No- Round 891 (DIV 3.A)

Problem Name: **Array Coloring**

Problem Link: <https://codeforces.com/contest/1857/problem/A>

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define yess cout<<"YES\n"
#define noo cout<<"NO\n"
#define newl cout<<endl
int main()
{
    _easy_
    int t;
    cin>>t;

```

```

while(t--)
{
    int n;
    cin>>n;
    int even=0,odd=0;
    for(int i=1;i<=n;i++)
    {
        int a;
        cin>>a;
        if(ev(a))
        {
            even++;
        }
        else
        {
            odd++;
        }
    }
    if(ev(odd))
    {
        yess;
    }
    else
    {
        noo;
    }
}
}

```

Output:

217639241	Aug/07/2023 20:53 ^{UTC+6}	Arsenic33	A - Array Coloring	GNU C++20 (64)	Accepted	15 ms	0 KB
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Problem No- Round 891 (DIV 3.B)

Problem Name: **Array Coloring**

Problem Link: <https://codeforces.com/contest/1857/problem/A>

Code:

```

#include<bits/stdc++.h>
using namespace std;
#define _easy_ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
int f(int x){if(x==1){return 1;}return x*f(x-1);}
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define vary(A,x,y,n) transform(A.begin()+x,A.begin()+y,A.begin()+x,[](int a){return a+n;})
#define newl cout<<endl
int main()
{
    _easy_
    int t;
    cin>>t;
    while(t--)
    {

```



```

str r;
cin>>r;
int b=-1;
for(int i=r.size()-1;i>=0;i--)
{
    if(r[i]>='5')
    {
        if(i!=0&& r[i-1]<'9')
        {
            r[i-1]++;
        }
        b=i;
        r[i]='0';
    }
}
if(b==0)
{
    cout<<1;
    for(int i=0;i<r.size();i++)
    {
        cout<<0;
    }
}
else if(b!=-1)
{
    for(int i=0;i<b;i++)
    {
        cout<<r[i];
    }
    for(int i=b;i<r.size();i++)
    {
        cout<<0;
    }
}
else
{
    cout<<r;
}
endl;
}
}

```

Output:

218295312	Aug/11/2023 18:26 ^{UTC+6}	Arsenic33	B - Maximum Rounding	GNU C++20 (64)	Accepted	31 ms	500 KB
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